

CAMERA CONTROLLER – signal flow

GND is a global net on every sheet. Double-click a block to open it.

1. Battery 2S + protection



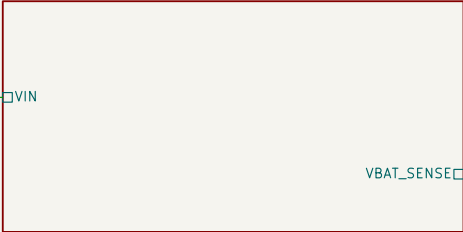
File: sheets/power_in.kicad_sch

2. Buck to 5 V



File: sheets/buck_5v.kicad_sch

3. Pack voltage sense



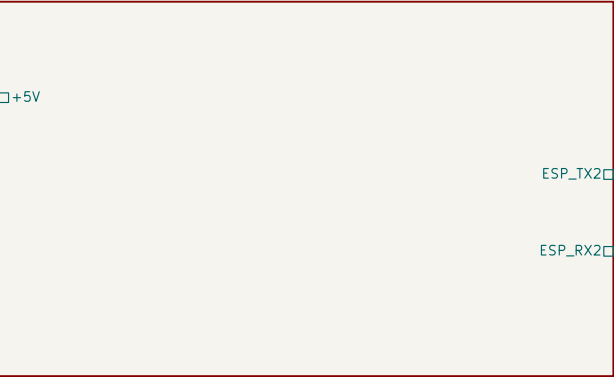
File: sheets/sense.kicad_sch

4. ESP32 socket



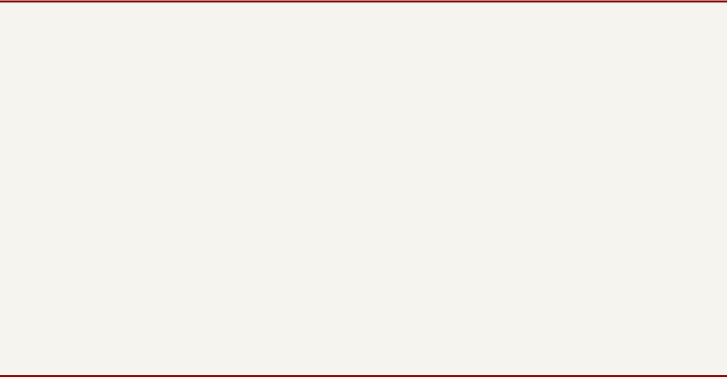
File: sheets/esp32.kicad_sch

5. RunCam Split 4



File: sheets/camera.kicad_sch

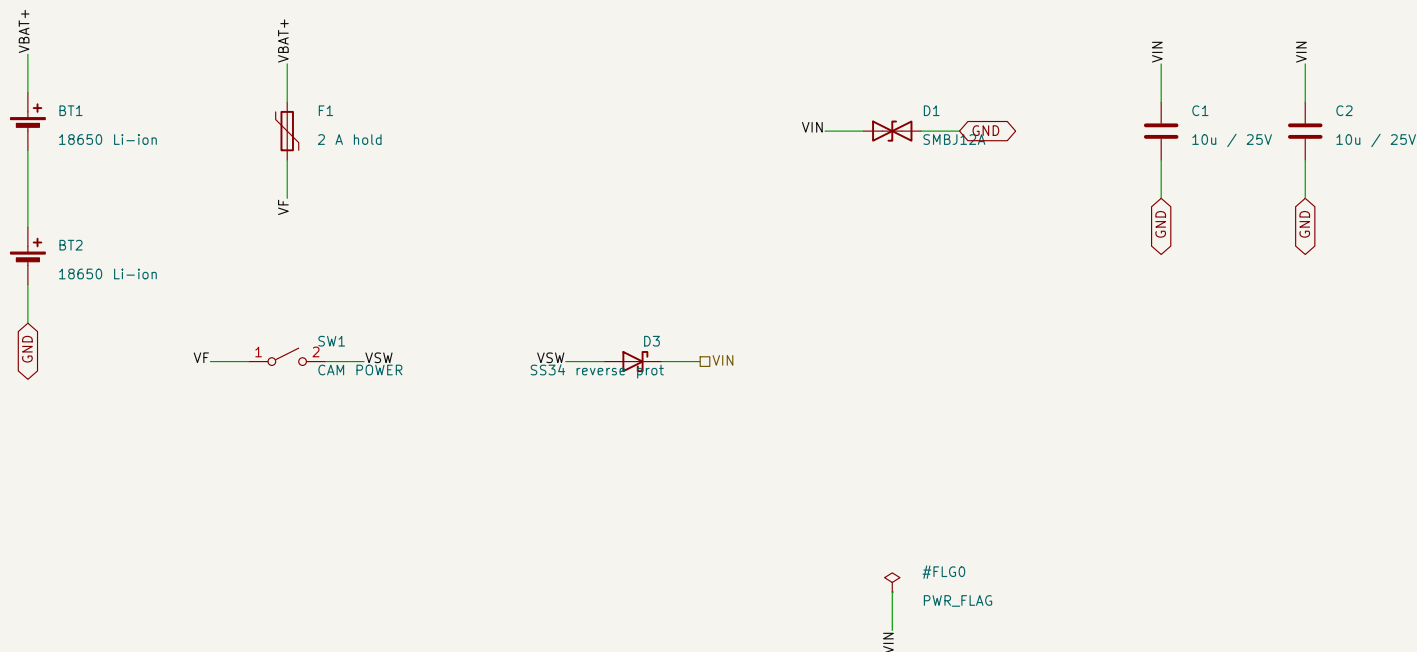
6. COTS / SRAD supplies



File: sheets/supplies.kicad_sch

Block 6 is electrically isolated from the rest of this board on purpose.

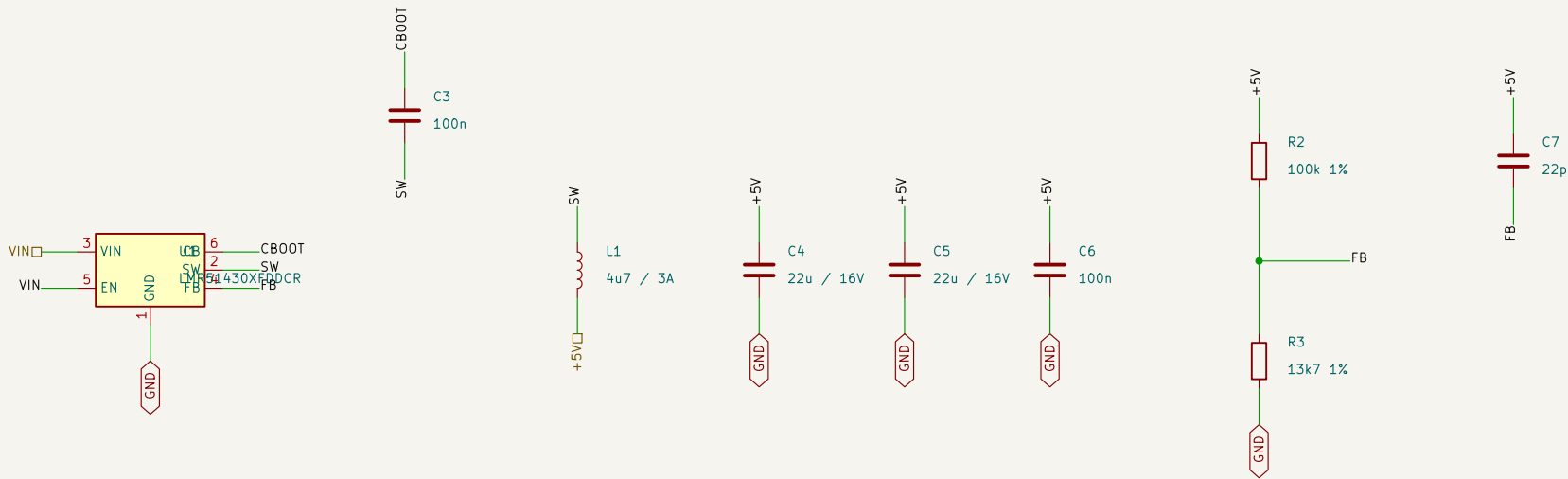
2 x 18650 in series -> 7.4 V nom, 8.4 V max



Power flag – ERC only.

Stanford SSI		
Sheet: /1. Battery 2S + protection/		
File: power_in.kicad_sch		
Title: 1. Camera + ESP32 supply (2S)		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 2/7

LMR51430 $V_{out} = 0.6 \times (1 + 100k / 13k7) = 4.98 \text{ V}$



Power flags – ERC only, not real parts.

Stanford SSI

Sheet: /2. Buck to 5 V/
File: buck_5v.kicad_sch

Title: 2. Buck 8.4 V -> 5.0 V

Size: A4

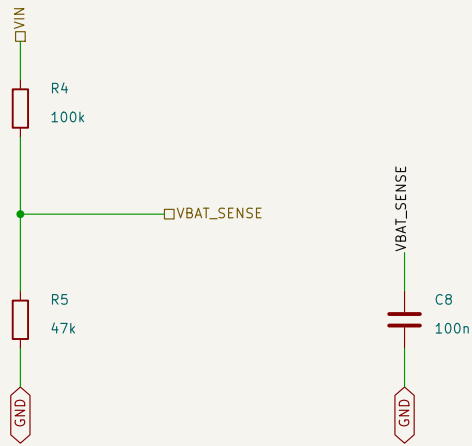
Date:

KiCad E.D.A. 10.0.0

Rev: A

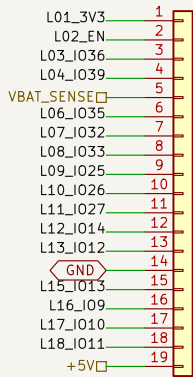
Id: 3/7

8.4 V x 47 / 147 = 2.69 V at the ADC

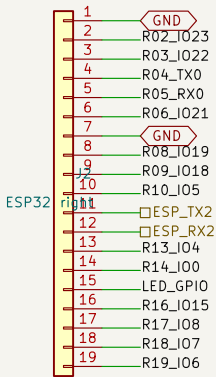


Stanford SSI		
Sheet: /3. Pack voltage sense/		
File: sense.kicad_sch		
Title: 3. Pack voltage sense		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0	Id: 4/7	

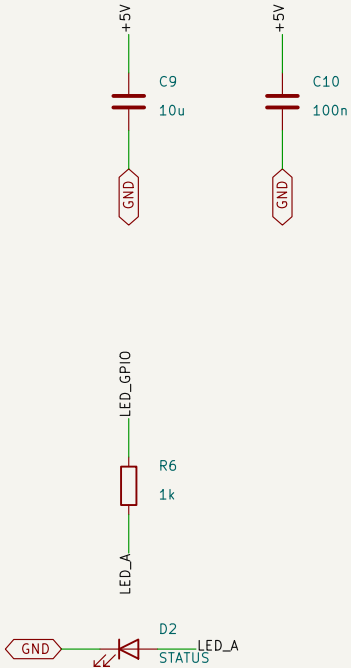
2 x 1x19 female header. GPIO6–11 go to the module SPI flash – do not use.



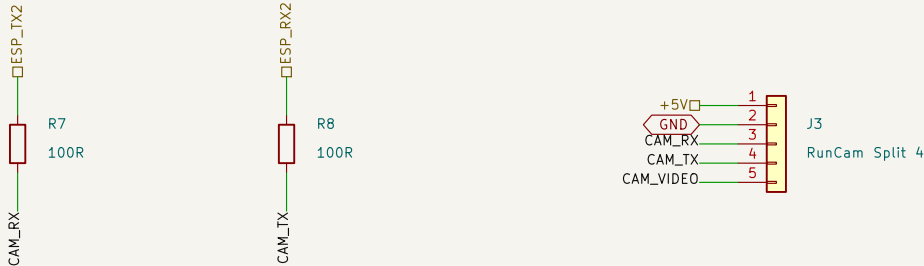
J1
ESP32 left



ESP32 right



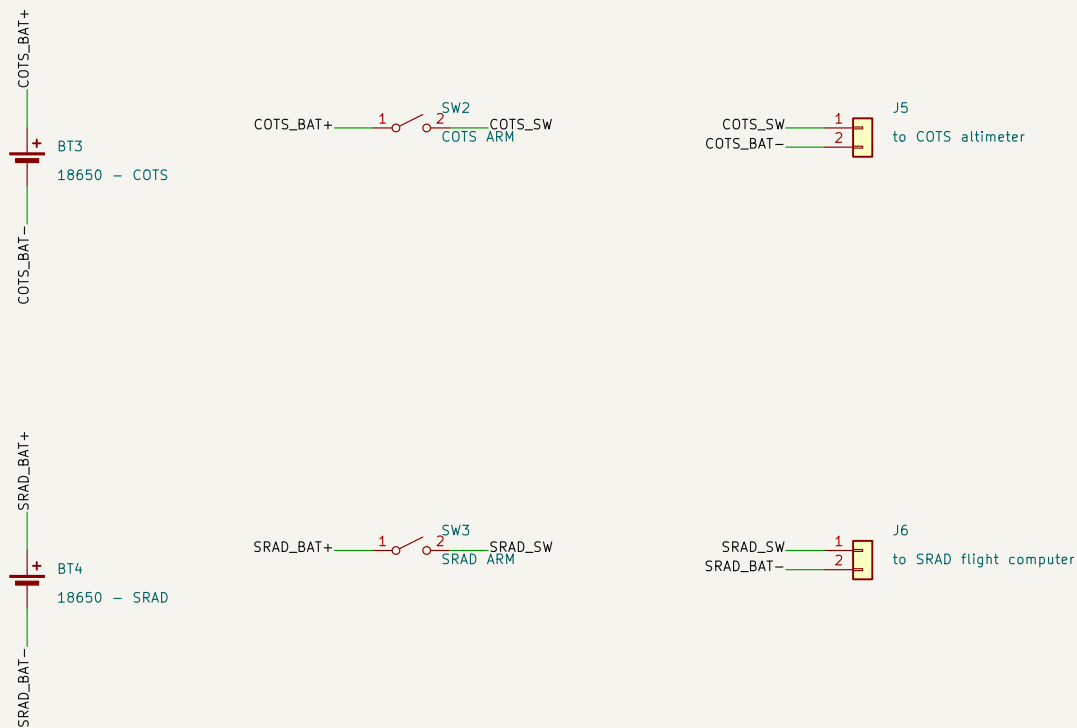
5–20 V in. Single 5–pin harness: power, ground and UART.



Stanford SSI		
Sheet: /5. RunCam Split 4/		
File: camera.kicad_sch		
Title: 5. RunCam Split 4		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 6/7

Own cell, own switch, per DTEG 6.9.1. SW2 / SW3 take the pin-switch leads; they sit next to the pin-switch stack. All connectors latch (Micro-Fit 3.0).

These returns are deliberately NOT tied to GND – that independence is the point.



Stanford SSI		
Sheet: /6. COTS / SRAD supplies/ File: supplies.kicad_sch		
Title: 6. Independent 1S supplies		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 7/7